Discrete Mathematics

Subject: Mathematics | NoteShare Library

DISCRETE MATHEMATICS

1. LOGIC

NOT: p
Implication: p -> q
Biconditional: p <-> q

Truth tables, De Morgan’s laws

2. SETS

Difference: A-B
Complement: A’

3. RELATIONS
Reflexive: aRa

Equivalence: all three

4. COMBINATORICS
P(n,r) = n!/(n-r)!
C(n,r) = n!/(r!(n-r)!)
Pigeonhole principle

5. GRAPH THEORY
Vertices V, Edges E
Degree sum = 2|E|
Euler: all even degrees
Hamiltonian: visits all once

6. RECURRENCE
T(n) = 2T(n/2) + n
Master theorem